

I am designing a research framework called HFPO (Heterogeneous Federated Prompt Optimization) for medical question answering using Large Language Models.

The core idea is not federated fine-tuning of model weights. Instead, hospitals collaborate to optimize system prompts while keeping patient data local.

The proposed workflow is:

A central server maintains a population of approximately 20 candidate prompts. The server broadcasts the same prompt population to multiple hospitals. Each hospital evaluates every prompt only on its own private dataset and returns only evaluation metrics (e.g., accuracy), never patient data. The server aggregates these scores into a per-prompt fitness representation (one score per hospital). A Genetic Algorithm performs parent selection (e.g., tournament selection and elitism). Instead of handcrafted mutation and crossover, a reasoning LLM generates new prompts conditioned on the selected parents and their evaluation feedback. The new prompt population is evaluated again across hospitals, repeating for multiple generations. The final outputs include globally robust prompts and hospital-specific high-performing prompts.

I would like to understand:

Has this exact problem of federated prompt optimization (rather than federated model training) already been studied? Are there papers combining Genetic Algorithms + LLM-based prompt generation + federated evaluation? What is the closest prior work? (e.g., EvoPrompt, PromptBreeder, OPRO, federated prompt learning, prompt tuning, federated prompt tuning) Which aspects of this framework appear novel, and which aspects have already been explored? Would this likely be considered a meaningful methodological contribution if supported by strong experiments? Are there publications specifically addressing prompt optimization under heterogeneous (non-IID) federated data distributions? What baselines would reviewers expect when evaluating such a framework?

Please provide references to peer-reviewed papers and identify any gaps in the literature that this framework could potentially address.

Heterogeneous Federated Prompt Optimization for Medical QA: How Novel Is It?

HFPO sits at the intersection of federated prompt learning, evolutionary prompt search, and medical LLM evaluation. Existing work gets close to each piece separately, but does not appear to combine them in exactly the way you propose.

Closest Existing Directions

Federated Prompt Learning / Tuning

- FedBPT, Fed-BBPT, FedPrompt, PromptFL, FedDTPT, DP-FPL, Probabilistic FL prompt tuning, SecFPP, FedPSF-LLM all let clients collaboratively train prompts (often *soft* prompts) with frozen PLMs/LLMs, focusing on communication efficiency, privacy, personalization, differential privacy, and robustness to **non-IID** data (Sun et al., 2023; Li et al., 2024; Zhao et al., 2022; Lin et al., 2023; Wu et al., 2024; Tran et al., 2025; Jiang et al., 2025; Weng et al., 2025; Hou et al., 2025).
- Several explicitly handle heterogeneous / non-IID data and imbalanced settings, often with specialized aggregation or personalization (Wu et al., 2024; Jiang et al., 2025; Weng et al., 2025; Hou et al., 2025).

These are strong prior art for “federated prompt learning,” but they generally optimize prompt *parameters* with gradient(-free) methods, not an explicit **population of discrete system prompts** evaluated only via black-box scores across sites.

Evolutionary + LLM-Based Prompt Optimization

- **EvoPrompt** uses an EA-style population of discrete prompts; a GA loop with selection plus LLM-based crossover/mutation, evaluated on a dev set, without using gradients (Guo et al., 2023).
- **GAPO** similarly applies genetic algorithms for prompt optimization, integrating multiple prompt generation strategies within an evolutionary framework (S'echeresse et al., 2025).

These give direct precedent for **GA + LLM-based prompt generation**, but in a centralized, single-dataset setting.

Federated, Non-IID Medical NLP

- Multiple works show federated fine-tuning or FL approaches on heterogeneous clinical text (e.g., layer-skipping FL for healthcare NLP; FL evaluation on biomedical IE) (Zhang & Li, 2025; Peng et al., 2024), and surveys highlight FL as a key privacy tool for biomedical LLMs (Wang et al., 2025).
- Med-PaLM 2 and FedMME show high-stakes medical QA and multimodal medical federation but not federated prompt optimization itself (Singhal et al., 2025; Wang et al., 2025).

What Seems Novel vs. Covered

Already Explored Elements

- **Federated prompt learning/tuning** (soft or discrete) with non-IID data and privacy constraints (Sun et al., 2023; Zhao et al., 2022; Lin et al., 2023; Wu et al., 2024; Jiang et al., 2025; Weng et al., 2025; Hou et al., 2025).
- **Prompt personalization** in federated settings (global + client-specific prompts) (Tran et al., 2025; Weng et al., 2025; Hou et al., 2025).
- **Evolutionary / GA + LLM-based prompt generation** for centralized prompt optimization (Guo et al., 2023; S'echeresse et al., 2025; Guo et al., 2023).
- **Black-box, gradient-free prompt optimization in federated LLM settings** (e.g., FedBPT/Fed-BBPT, FedDTPT) (Sun et al., 2023; Li et al., 2024; Lin et al., 2023; Wu et al., 2024).

Aspects That Appear Novel

Based on the abstracts, there does **not** appear to be:

- A framework that:
 - Maintains a **global population of discrete system prompts** on a server,
 - **Broadcasts the same population** to multiple clients,
 - Lets each client **only return evaluation scores per prompt**,
 - Aggregates these into a **per-prompt, per-client fitness vector**, and
 - Uses a **GA where LLM(s) implement mutation/crossover** conditioned on those multi-hospital fitness signals,
 - To jointly produce **globally robust and site-specific prompts** for medical QA.

This specific combination — **EvoPrompt/GAPO-style LLM-driven evolutionary search + purely score-based federated evaluation across heterogeneous clinical sites** — does not show up in the retrieved work.

Non-IID Federated Prompt Optimization

- Several methods explicitly target non-IID or imbalanced settings in federated prompt learning, including **FedDTPT** (robustness to non-IID), **probabilistic federated prompt-tuning** for extreme heterogeneity, and personalized FPP methods like **SecFPP** (Wu et al., 2024; Weng et al., 2025; Hou et al., 2025).
- These typically aggregate prompt parameters or probabilistic representations, not *evaluation-only* fitness in a black-box evolutionary loop.

Likely Contribution & Reviewer Expectations

Methodological Contribution

If implemented and evaluated rigorously, HFPO would likely be viewed as a **non-trivial methodological contribution**, because:

- It introduces a **new optimization paradigm**: federated, black-box, population-based prompt search where **no model parameters or prompts are shared as gradients/weights**, only scores.
- It connects **EvoPrompt-style prompt evolution** with **federated evaluation over non-IID medical data**, which current works only cover separately (Wu et al., 2024; Guo et al., 2023; S'echeresse et al., 2025; Weng et al., 2025; Hou et al., 2025).

Strength would depend on showing:

- Clear **privacy and communication advantages** over federated prompt tuning / federated fine-tuning baselines (Sun et al., 2023; Li et al., 2024; Zhao et al., 2022; Lin et al., 2023; Zhang & Li, 2025; Jiang et al., 2025; Weng et al., 2025; Hou et al., 2025).
- Gains in **cross-hospital robustness** and/or **personalization** on realistic medical QA benchmarks (Singhal et al., 2025; Wang et al., 2025; Peng et al., 2024; Wang et al., 2025).

Relevant Baselines

Reviewers would likely expect comparisons to:

Baseline Type	Example Methods	Citations
Centralized prompt optimization (single dataset)	EvoPrompt, GAAPO	(Guo et al., 2023; S'echeresse et al., 2025; Guo et al., 2023)
Federated prompt tuning (global prompt)	FedBPT, Fed-BBPT, FedPrompt, PromptFL	(Sun et al., 2023; Li et al., 2024; Zhao et al., 2022; Lin et al., 2023; Guo et al., 2022)
Federated prompt personalization	pFedPrompt, pFedPG, FedOTP, FedPGP, SecFPP, DP-FPL, probabilistic FL prompt tuning	(Tran et al., 2025; Weng et al., 2025; Hou et al., 2025)
Federated fine-tuning baselines in healthcare	Layer-skipping FL, FedAvg/FedProx setups from biomedical FL studies	(Zhang & Li, 2025; Peng et al., 2024)

Baseline Type	Example Methods	Citations
NLP		
Non-federated strong medical QA LLMs	Med-PaLM 2–style centralized fine-tuned models or strong few-shot prompting	(Singhal et al., 2025; Peng et al., 2024)

FIGURE 1 Key baseline families for HFPO evaluation

Gaps HFPO Could Address

- **No current work** combining: GA + LLM-based **discrete prompt evolution**, **federated**, **score-only evaluation** across heterogeneous institutions, and **explicit multi-objective selection** for global vs. site-specific prompts (Sun et al., 2023; Lin et al., 2023; Wu et al., 2024; Guo et al., 2023; S'echeresse et al., 2025; Weng et al., 2025; Hou et al., 2025).
- Limited exploration of **federated prompt optimization specifically for medical QA**, where question distributions and clinical styles vary strongly across hospitals (Singhal et al., 2025; Wang et al., 2025; Peng et al., 2024; Wang et al., 2025).
- Underexplored trade-offs between **privacy (score-only sharing)**, **communication cost**, and **robustness to extreme non-IID** in prompt-only frameworks (Sun et al., 2023; Li et al., 2024; Lin et al., 2023; Wu et al., 2024; Weng et al., 2025; Hou et al., 2025).

Conclusion

Existing research already covers federated prompt learning (including non-IID and personalization) and evolutionary / GA + LLM-based prompt optimization, but in **separate** lines of work. There is no clear evidence of a method that unifies them into a score-only, GA-driven, federated prompt evolution framework for medical QA. HFPO therefore appears to target a real gap, and with strong experiments against modern federated prompt and fine-tuning baselines, it would likely be viewed as a meaningful methodological advance.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

- Guo, Q., Wang, R., Guo, J., Li, B., Song, K., Tan, X., Liu, G., Bian, J., Yang, Y., University, T., & Research, M. (2023). EvoPrompt: Connecting LLMs with Evolutionary Algorithms Yields Powerful Prompt Optimizers.
- Guo, Q., Wang, R., Guo, J., Li, B., Song, K., Tan, X., Liu, G., Bian, J., & Yang, Y. (2023). Connecting Large Language Models with Evolutionary Algorithms Yields Powerful Prompt Optimizers. *ArXiv, abs/2309.08532*. <https://doi.org/10.48550/arxiv.2309.08532>
- Guo, T., Guo, S., Wang, J., Tang, X., & Xu, W. (2022). PromptFL: Let Federated Participants Cooperatively Learn Prompts Instead of Models – Federated Learning in Age of Foundation Model. *IEEE Transactions on Mobile Computing*, 23, 5179–5194. <https://doi.org/10.1109/tmc.2023.3302410>
- Hou, S., Li, S., & Buyukates, B. (2025). Privacy-preserving Prompt Personalization in Federated Learning for Multimodal Large Language Models. *ArXiv, abs/2505.22447*. <https://doi.org/10.48550/arxiv.2505.22447>

- Jiang, Y., Li, Z., & Song, B. (2025). Fine-tuning large language models in federated learning with fairness-aware prompt selection. *Neural networks : the official journal of the International Neural Network Society*, 194, 108160. <https://doi.org/10.1016/j.neunet.2025.108160>
- Li, Y., Sun, J., Liu, Y., Zhang, Y., Li, A., Chen, B., Roth, H. R., Xu, D., Chen, T., & Chen, Y. (2024). Federated Black-box Prompt Tuning System for Large Language Models on the Edge. *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking*. <https://doi.org/10.1145/3636534.3698856>
- Lin, Z., Sun, Y., Shi, Y., Wang, X., Huang, L., Shen, L., & Tao, D. (2023). Efficient Federated Prompt Tuning for Black-box Large Pre-trained Models. *ArXiv, abs/2310.03123*. <https://doi.org/10.48550/arxiv.2310.03123>
- Peng, L., Luo, G., Zhou, S., Chen, J., Xu, Z.-C., Sun, J., & Zhang, R. (2024). An in-depth evaluation of federated learning on biomedical natural language processing for information extraction. *NPJ Digital Medicine*, 7. <https://doi.org/10.1038/s41746-024-01126-4>
- S'echeresse, X., Guilbert--Ly, J.-Y., & De Torcy, A. V. (2025). GAPO: genetic algorithmic applied to prompt optimization. *Frontiers in Artificial Intelligence*, 8. <https://doi.org/10.3389/frai.2025.1613007>
- Singhal, K., Tu, T., Gottweis, J., Sayres, R., Wulczyn, E., Amin, M., Hou, L., Clark, K., Pfohl, S. R., Cole-Lewis, H., Neal, D., Rashid, Q., Schaekermann, M., Wang, A., Dash, D., Chen, J. H., Shah, N. H., Lachgar, S., Mansfield, P., . . . Natarajan, V. (2025). Toward expert-level medical question answering with large language models. *Nature Medicine*, 31, 943 - 950. <https://doi.org/10.1038/s41591-024-03423-7>
- Sun, J., Xu, Z., Yin, H., Yang, D., Xu, D., Chen, Y., & Roth, H. R. (2023). FedBPT: Efficient Federated Black-box Prompt Tuning for Large Language Models. 47159-47173. <https://doi.org/10.48550/arxiv.2310.01467>
- Tran, L., Sun, W., Patterson, S., & Milanova, A. (2025). Privacy-Preserving Personalized Federated Prompt Learning for Multimodal Large Language Models. *ArXiv, abs/2501.13904*. <https://doi.org/10.48550/arxiv.2501.13904>
- Wang, C., Li, M., He, J., Wang, Z., Darzi, E., Chen, Z., Ye, J., Li, T.-X., Su, Y.-C., Ke, J., Qu, K., Li, S., Yu, Y., Liò, P., Wang, T., Wang, Y. G., & Shen, Y. (2025). A survey for large language models in biomedicine. *Artificial intelligence in medicine*, 170, 103268. <https://doi.org/10.1016/j.artmed.2025.103268>
- Wang, N., Deng, Y., Fan, S., Yin, J., & Ng, S.-K. (2025). Multi-Modal One-Shot Federated Ensemble Learning for Medical Data with Vision Large Language Model. *ArXiv, abs/2501.03292*. <https://doi.org/10.48550/arxiv.2501.03292>
- Weng, P.-Y., Hoang, M., Nguyen, L. M., Thai, M. T., Weng, T.-W., & Hoang, T. (2025). Probabilistic Federated Prompt-Tuning with Non-IID and Imbalanced Data. *ArXiv, abs/2502.19752*. <https://doi.org/10.48550/arxiv.2502.19752>
- Wu, J., Chen, S., Yang, Y., Li, Y., Hou, S., Jing, R., Wang, Z., Chen, W., & Tian, Z. (2024). FedDTPT: Federated Discrete and Transferable Prompt Tuning for Black-Box Large Language Models. *ArXiv, abs/2411.00985*. <https://doi.org/10.48550/arxiv.2411.00985>
- Zhang, L., & Li, Y. (2025). Federated Learning with Layer Skipping: Efficient Training of Large Language Models for Healthcare NLP. *ArXiv, abs/2504.10536*. <https://doi.org/10.48550/arxiv.2504.10536>
- Zhao, H., Du, W., Li, F., Li, P., & Liu, G. (2022). FedPrompt: Communication-Efficient and Privacy-Preserving Prompt Tuning in Federated Learning. *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 1-5. <https://doi.org/10.1109/icassp49357.2023.10095356>